AN IRRATIONALITY MEASURE FOR CATALAN'S CONSTANT

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Abstract. Let $G = \sum_{j\geq 0}(-1)^j(2j+1)^{-2}$ be Catalan's constant. Building on OpenAI's recent proof that G is irrational, we prove that $|G - a/b| > b^{-728}$ for every integer a and every sufficiently large integer b; thus the irrationality exponent of G is at most 728. The irrationality proof compares, for a determinant Δ whose entries are linear forms in 1 and G, a lower bound coming from the denominators of Δ with an upper bound coming from an integral representation. To obtain a measure we regard the determinant as a polynomial $\Delta_N(X)$ in the constant and bound it, not only at $X = G$, but on an exponentially small neighbourhood of G. We control the derivative of the determinant through a Gauss quadrature formula, which turns the perturbation into point evaluations of the row functions, and an identity of Andréief type with fixed points. We use a determinant of size $20N$ rather than $48N$, and we show that every coefficient of the resulting integer polynomial is divisible by large powers of the primes in three intervals, by proving rank deficiencies of the reduced matrix modulo these primes with Chebyshev identities in characteristic p. Prime squares in the denominator of a rational approximation are recovered from the coefficients, which removes the cost of selecting a good prime. The energy estimates at the real place use trial potentials with complex poles placed near the edges of the support of the extremal measure. All steps are formalized in the Lean proof assistant, with only the standard axioms.

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1. Introduction

Catalan's constant

$$G = \sum_{j=0}^{\infty} \frac{(-1)^j}{(2j+1)^2} = L(2, \chi_{-4}) = 0.91596\,55941\,77219\ldots$$

is the value at $s = 2$ of the Dirichlet L-function of the nontrivial character modulo 4. Its arithmetic has long been studied by analogy with $\zeta(2)$ and $\zeta(3)$, but until recently it was not known to be irrational. In October 2026 OpenAI [25] proved that G is irrational and verified the proof in Lean. (An earlier claimed proof [33] was shown to contain a gap [36].)

The *irrationality exponent* $\mu(\xi)$ of a real number ξ is the infimum of the real numbers μ such that $|\xi - a/b| > b^{-\mu}$ for all integers a and all sufficiently large integers b. Every irrational number has $\mu(\xi) \geq 2$, and almost every real number has $\mu(\xi) = 2$. An *irrationality measure* is an explicit upper bound for $\mu(\xi)$.

Theorem 1.1. *For every integer a and every sufficiently large integer b,*

$$\left| G - \frac{a}{b} \right| > b^{-728}.$$

In particular $\mu(G) \leq 728$.

The proof uses the prime number theorem. The theorem and every result in this paper have been verified in the Lean 4 proof assistant on top of OpenAI's formalization, with no unproved steps and only the standard axioms (Section 9).

1.1. **Previous results.** Irrationality proofs for the values of L-functions at integers follow, almost without exception, the pattern of Apéry's proof for $\zeta(3)$ [1]. One constructs integer linear forms $q_n\xi - p_n$ that tend to zero, with $q_n \neq 0$, faster than the growth of their coefficients would allow for a rational ξ. If $|q_n\xi - p_n| = e^{-\alpha n + o(n)}$ and $|q_n| = e^{\beta n + o(n)}$, then ξ is irrational as soon as $\alpha > 0$, and $\mu(\xi) \leq 1 + \beta/\alpha$. Beukers [2] gave the linear forms for $\zeta(2)$ and $\zeta(3)$ as integrals of Legendre-type polynomials. Hata [14] refined such integrals and introduced the systematic use of the primes that divide all the coefficients of the forms, an idea going back to Chudnovsky. Rhin and Viola [28, 29] found that families of integrals for $\zeta(2)$ and $\zeta(3)$ are permuted by a finite group, and used the invariance of the integrals to divide all their coefficients by a large common factor. This *arithmetic method* gave the sharpest measures for $\zeta(2)$ and $\zeta(3)$, and Marcovecchio [24] applied it to $\log 2$. For π, Zeilberger and Zudilin [37] obtained $\mu(\pi) \leq 7.1032\ldots$ from Salikhov's integrals [32] with an additional arithmetic saving.

Since September 2026 a series of new results in this area has appeared, several of them obtained with AI systems. OpenAI proved $\mu(\pi) = 2$, with a Lean formalization [26], and Liu, Jiang and Zhang proved $\mu(\log r) = 2$ for every rational $r > 0$, $r \neq 1$ [23]. Giraudin lowered the bounds for $\mu(\zeta(2))$ and $\mu(\zeta(3))$ to 5.0193784 and 5.5138800 by applying the Rhin–Viola method at new integral directions [12, 13]. Leibtag, Barkan, Kalisch, Weinbaum, Shalyt and Kaminer [21] showed that the classical constructions are one-dimensional trajectories in conservative matrix fields, and by searching over other trajectories obtained $\mu(\zeta(2)) < 4.62675$ and $\mu(\zeta(3)) < 5.4197$. Prellberg [27] proved that $\mathrm{Li}_2(-1/4)$, $\mathrm{Li}_2(1/5)$ and $\mathrm{Li}_2(-1/3)$ are irrational with the integrals of Rhin–Viola and Viola–Zudilin. Fauzan announced a proof that $\zeta(5)$ is irrational, using Hankel determinants [10], and Lai combined this method with simultaneous approximations to show that 1, $\zeta(3)$ and $\zeta(4)$ are linearly independent over $\mathbb{Q}$ [19]. Several of these arguments, like the present one, use determinants of large size rather than single sequences

of linear forms; Zudilin's determinantal criterion [40] is an earlier instance of this idea.

For Catalan's constant the analogous linear forms exist but do not decay fast enough. Rivoal and Zudilin [30] proved that at least one of the numbers $\beta(2), \beta(4), \dots, \beta(2m)$ is irrational for suitable m, where β is the Dirichlet beta function; $G = \beta(2)$ is not singled out. Zudilin [39] found an Apéry-like recurrence for G whose solutions give linear forms that converge to zero, but too slowly to prove irrationality, and [38, 17, 18, 41] describe related constructions and their limits. Eskandari, Murty and Nemoto [9] interpret linear forms in Catalan's constant through mixed motives. Leibtag et al. [21] found further Apéry-like approximations to G in their matrix fields; the best they report falls just short of proving irrationality.

OpenAI's argument [25] has a different shape. Instead of a sequence of linear forms it uses a determinant Δ of size $48N$ whose entries are rational linear combinations of 1 and G. Determinants of linear forms were used for irrationality in Zudilin's determinantal approach [40]; here the entries are double integrals against the Cauchy kernel $1/(1-ts)$. If G were rational, the denominators of the entries would give a lower bound $|\Delta| \geq e^{-(\mathsf{A}+o(1))n^2}$ with $n = 48N$. An integral representation, of the kind that appears in random matrix theory through Andréief's identity [11], gives the upper bound $|\Delta| \leq e^{-(\mathsf{U}+o(1))n^2}$, and an energy estimate shows $\mathsf{U} > \mathsf{A}$. The margin is small: $\mathsf{U} - \mathsf{A} \approx 1.3 \cdot 10^{-4}$. The nonvanishing of Δ along the primes $N = p$ is proved separately by reduction modulo p.

The first irrationality measure for G was obtained by Jha [15], who proved $\mu(G) \leq 10^8$. Carreira [6] studied general five-parameter determinant designs, obtained $\mu(G) < 2294$ for one of them, and formulated the best real-place constant of the method as an equilibrium problem for two measures; in [5] he records attempts to carry the method to $\beta(4)$. Eskandari [8] studies rational approximations to G from another point of view. For $L(2, \chi_{-3})$, Calegari, Dimitrov and Tang proved linear independence of 1, $\zeta(2)$ and $L(2, \chi_{-3})$ [3], and their holonomy bounds give $\mu(L(2, \chi_{-3})) \leq 24781$ [4, Thm. 5.1]. A companion paper [22] proves measures for $G \pm 5\pi^2/48$ and $L(2, \chi_{-3})$ with a related method.

1.2. **Outline of the proof.** Section 2 explains the method and the new ideas without technicalities. Section 3 recalls OpenAI's construction, and Section 4 defines the determinant of size $20N$ and proves its degree bound, its denominator bound and its nonvanishing. Section 5 proves the divisibility of the coefficients, and Section 6 contains the estimates at the real place. Section 7 proves Theorem 1.1, Section 8 discusses the limits of the method, and Section 9 describes the formalization; the disclosure of AI use follows it.

1.3. **Notation and conventions.** We use the notation of OpenAI's paper [25] for the construction and refer to it for details; citations of the form [25,

Prop. 2.1] point to its numbered statements. We write log for the natural logarithm, v_p for the p-adic valuation, $\theta(x) = \sum_{p \le x} \log p$ for Chebyshev's function, T_k and U_k for the Chebyshev polynomials of the first and second kind, and $y_+ = \max(y, 0)$.

2. Overview of the method and of the new ideas

2.1. **From irrationality to a measure.** Write OpenAI's matrix as $\mathcal{A}_N = \mathcal{A}_N^0 + G\,\mathcal{L}_N$ with rational matrices $\mathcal{A}_N^0$ and $\mathcal{L}_N$, and put $\Delta_N(X) = \det(\mathcal{A}_N^0 + X\mathcal{L}_N)$, a polynomial in the constant. A symmetry of the rows of $\mathcal{L}_N$ gives a degree bound $\deg \Delta_N \le dN$, and multiplication by a common denominator and a factorial gives an integer polynomial Q_N.

Two things are needed to turn OpenAI's argument into a measure. First, the hypothesis $G = a/b$ must be replaced by a rational approximation: the determinant must be small not only at $X = G$ but at every X within $e^{-\rho N}$ of G, for some constant ρ. Second, the nonvanishing at a rational point a/b must hold uniformly, along primes p not dividing b, and the denominator bound must be uniform in a/b. Given these, a standard argument (Proposition 7.1) gives $\mu(G) \le d\rho/\eta$ up to a factor coming from the choice of the prime, where ηN^2 is the margin between the two bounds. Making ρ small and η large is the content of the paper.

We use OpenAI's construction with different parameters, which gives a determinant of size $20N$ and degree at most $18N$ (Section 4). Its denominators are computed exactly at every prime, including the contribution of linear relations among the rows modulo p, and its nonvanishing is proved by reduction modulo p under the weaker hypothesis $p^2 \nmid b$.

2.2. **New idea 1: a perturbation bound by quadrature.** To bound $\Delta_N(x) - \Delta_N(G)$ for x near G, we write the matrix $\mathcal{L}_N$ by an exact Gauss-type quadrature formula as a combination of point evaluations of the row functions (Lemma 6.2). Expanding the determinant column by column gives mixed determinants in which some columns are integrals and others are point values. An identity of Andréief type with fixed points (Lemma 6.4) writes each of them as an integral over fewer variables, and the energy method bounds the integrand. The quadrature weights are bounded on a circle slightly larger than the unit circle, with nodes chosen away from the point where the energy fields are largest. This gives a perturbation exponent ρ of the size of the constants in the energy estimate.

2.3. **New idea 2: divisibility of all coefficients.** The core of the arithmetic is that every coefficient of Q_N is divisible by a power of each prime p in three intervals, $(52N/9, 44N/7)$, $(8N, 9N)$ and $(4N, 9N/2)$, given by the rank deficiency of the reduced matrix $p^2\mathcal{A}_N^0 \bmod p$ (Section 5). We prove these rank deficiencies with identities for Chebyshev polynomials in characteristic p, such as $T_p = X^p$ and $U_{p-1} = (X^2-1)^{(p-1)/2}$, and their consequences for the digits in base p of the row polynomials. By the prime

number theorem the common factor has logarithm at least $0.649N^2$, which is added to the margin η. This is in the spirit of the arithmetic method of Hata and Rhin–Viola, but the common factor comes from linear algebra modulo p rather than from a group of symmetries.

2.4. **New idea 3: prime squares in the denominator.** The polynomial $Q_N(X/q^2)$ is still integral at the primes $N < q \le 2N$, so q^{2j} divides the j-th coefficient of Q_N. If b is divisible by the squares of many such primes, the integer $b^{\deg Q_N} Q_N(a/b)$ is divisible by their product. Choosing the *largest* prime p with $p^2 \nmid b$ in a suitable interval, every larger prime of the interval has its square dividing b, and its contribution is recovered from the coefficients (Section 7). This removes the loss usually caused by the choice of the prime.

2.5. **New idea 4: trial potentials with complex poles.** The energy estimate needs explicit trial sequences, whose generating functions are potentials. The extremal measure of the energy problem has a gap around $x = 0$ in its support; near the edges of the gap, and near $x = 0$, Chebyshev expansions converge slowly. We add terms $\operatorname{Re} r \log(1 - 2xz + z^2)$, which are logarithmic potentials of point charges at complex points close to these edges (Section 6). The resulting fields are bounded above on their whole domains by a verified computation on intervals, with an exact local Taylor expansion of the polynomial part.

2.6. **The deduction.** Combining these, the degree is $d = 18$, the margin is $\eta = 64169/25000$ and the perturbation exponent is $\rho = 259299/2500$. Since the choice of the prime costs nothing, the exponent is the limit

$$\mu(G) \le \frac{18\rho}{\eta} = \frac{46673820}{64169} = 727.357\ldots,$$

which gives Theorem 1.1.

2.7. **Formal verification.** Every step, including the arithmetic of the denominators, the rank deficiencies, the energy estimates and their numerical bounds, and the prime number theorem, is formalized in Lean 4 with Mathlib [7, 34] on top of OpenAI's formalization. The written proofs below are at the level of a research article; for long computations we point to the formal proof, which is the basis for the correctness of Theorem 1.1.

3. OpenAI's construction

We recall the construction of [25] in the generality in which we use it. It depends on positive integers n, b, q, g, h and L, C, H, A, all proportional to N; OpenAI's choice is $n = 48N$, and ours is given in Section 4. For $0 \le r < n$ the row polynomials are

$$P_r(t) = (1-t)^h t^{C-1} T_{|r-g|}(1/t), \qquad \deg P_r < H, \tag{3.1}$$

together with companion polynomials D_r built from $U_{|r-g|-1}$ [25, (3)]. Thus P_r depends only on $|r-g|$. Put $f(t)=\sqrt{1-t^2}$ and $d\mu(t)=|t|\,dt/f(t)$ on $(-1,1)$, and

$$A_r(t)=P_r(t)-\frac{3}{2}\,\mathbf{1}_{\{t>0\}}\frac{f(t)}{t}D_r(t),\qquad \psi_k(s)=s^{b+k}(1-s)^q.$$

The determinant studied by OpenAI is that of the $n\times n$ matrix

$$\mathcal{A}_N(r,k)=\int_{-1}^{1}\int_{0}^{1}A_r(t)\,\frac{\psi_k(s)}{1-ts}\,ds\,d\mu(t)\qquad(0\le r,k<n). \tag{3.2}$$

The factor $(1-s)^q$ in ψ_k amounts to a finite difference of order q applied to the columns. The entries are rational linear combinations of 1 and G: there are rational matrices $\mathcal{A}_N^0$ and $\mathcal{L}_N$ with

$$\mathcal{A}_N=\mathcal{A}_N^0+G\,\mathcal{L}_N,\qquad \mathcal{L}_N(r,k)=\sum_{v=0}^{q}(-1)^v\binom{q}{v}\sum_i[t^i]P_r\cdot 4\,c_{(i-b-k-v)/2}, \tag{3.3}$$

where $c_l=4^{-l}\binom{2l}{l}$ for integers $l\ge 0$ and $c_z=0$ otherwise [25, Prop. 2.1 and (20)].

OpenAI's proof has three parts, and we use all of them.

The finite places. For every rational z and every prime ℓ, OpenAI bounds $v_\ell\det(\mathcal{A}_N^0+z\mathcal{L}_N)$ from below in terms of ℓ, N and the denominator of z [25, Prop. 2.2, §3]. The proof groups the coefficients of the row polynomials by their digits in base ℓ and uses the ℓ-adic valuations of the moments of μ and of the numbers c_l.

Nonvanishing. For a large prime p and $N=p$, the matrix $p^2(\mathcal{A}_p^0+z\mathcal{L}_p)$ has p-integral entries when $p\nmid \operatorname{den} z$, and its reduction modulo p does not depend on z and is invertible [25, Prop. 4.1]. Hence the determinant is nonzero and its p-adic valuation is known exactly. OpenAI states this under the hypothesis $G=z$, but the proof uses only the arithmetic of the rational matrix $\mathcal{A}_p^0+z\mathcal{L}_p$, and only the condition $p\nmid\operatorname{den} z$. This uniformity in z is what makes a measure possible.

The real place. The entries (3.2) are double integrals against the Cauchy kernel $1/(1-ts)$. By Andréief's identity the determinant is an integral over n points $t_i\in(-1,1)$ and n points $s_j\in(0,1)$ of a product of three determinants, one of which is the Cauchy determinant $\det[1/(1-t_is_j)]$. After the change of variables $t=2x/(1+x^2)$, OpenAI bounds the integrand by the exponential of n^2 times a discrete logarithmic energy of the points x_i and s_j [25, §5]. The maximum of this energy is bounded by a dual argument: two trial sequences are chosen, two squares are completed, and the resulting functions of one variable are bounded above by an explicit computation [25, Prop. 6.1, §7]. We describe this in Section 6, where we extend it.

4. The determinant of size $20N$

4.1. **The parameters.** We use OpenAI's construction with

$$n = 20N, \quad b = 3N, \quad q = N, \quad g = 2N, \quad h = N, \qquad L = 24N, \quad C = 26N, \quad H = 27N, \quad A = 8N,$$

and write

$$\Delta_N(X) = \det(\mathcal{A}_N^0 + X\mathcal{L}_N) \in \mathbb{Q}[X],$$

so that $\Delta_N(G) = \det \mathcal{A}_N$. These parameters were found by a search over OpenAI's family, scored by the margin between the constants at the real place and at the finite places; the search is not part of the proof.

4.2. **The degree.**

Lemma 4.1. $\deg \Delta_N \le 18N$.

Proof. By (3.3) the row r of $\mathcal{L}_N$ depends only on P_r, and hence only on $|r-g|$. So the rows r and $2g-r$ of $\mathcal{L}_N$ agree for $0 \le r < g$. Subtracting row r from row $2g-r$ for each $r < g$ does not change the determinant and makes g rows of the new coefficient of X vanish. A determinant $\det(B + XL')$ in which L' has $n-g$ nonzero rows has degree at most $n-g = 18N$ in X. □

4.3. **Rank deficiency and valuation.** We use repeatedly the following elementary fact.

Lemma 4.2. *Let M be an $n \times n$ matrix with entries in $\mathbb{Z}_{(p)}$ whose reduction modulo p has rank at most $n-e$. Then $v_p(\det M) \ge e$.*

Proof. Factor the reduction through $\mathbb{F}_p^{n-e}$ and lift the two factors to integral matrices U, V, so that $M = UV + pE$ with E integral at p. Expanding the determinant multilinearly in the columns, every term that uses more than $n-e$ columns of UV vanishes, and every other term has at least e factors p. □

4.4. **The denominators.** For an odd prime p with $27N < p^2$, the method of [25, §3], applied to the present parameters, gives

$$v_p(\Delta_N(z)) \ge -N\,\varphi(p/N)$$

for every rational z whose denominator is prime to p, where

$$\varphi(x) = \begin{cases} 40 & 0 \le x \le 10, \\ 60 - 2x & 10 \le x \le 12, \\ 36 & 12 \le x \le 27/2, \\ 63 - 2x & 27/2 \le x \le 14, \\ 91 - 4x & 14 \le x \le 16, \\ 75 - 3x & 16 \le x \le 24, \\ 27 - x & 24 \le x \le 27, \\ 0 & x \ge 27. \end{cases}$$

For each range of p/N, the function φ counts the columns whose entries have a pole at p, once the coefficients of the row polynomials are grouped by their digits in base p and the column filter is taken into account. The bound holds for the determinant itself and not only for a majorant of it.

Modulo p, some rows become linearly dependent, and by Lemma 4.2 this raises the valuation. For primes with $22N/3 < p < 26N/3$, three groups of rows satisfy an exact linear relation after reduction. The relation follows from the identities

$$T_p(Y) = Y^p, \qquad U_{p-1}(Y) = (Y^2-1)^{(p-1)/2} \qquad \text{in } \mathbb{F}_p[Y], \tag{4.1}$$

together with a rational identity between OpenAI's moment coefficients. This gives an additional saving $N\,\gamma(p/N)$, where

$$\gamma(x) = 3(x-22/3)_+ - 6(x-8)_+ + 3(x-26/3)_+$$

is supported on $[22/3, 26/3]$, where $\varphi = 40$.

We use the prime number theorem in the weighted form

$$\frac{1}{N}\sum_{p} F(p/N)\log p \longrightarrow \int_0^\infty F(x)\,dx \qquad (N\to\infty), \tag{4.2}$$

for bounded, Riemann integrable, compactly supported F. Since each prime contributes $N(\varphi-\gamma)(p/N)\log p$, the odd primes contribute

$$N^2\int_0^{27}(\varphi-\gamma)(x)\,dx = \frac{8795}{4800}(20N)^2$$

to the logarithm of the common denominator, up to $o(N^2)$; the primes $p \le \sqrt{27N}$ contribute $o(N^2)$.

At the prime 2 the bound comes from the expansion of the determinant into minors. Its exponent is governed by a quadratic form in the numbers m, ℓ of columns of two exceptional types, and the identity

$$\begin{aligned}&\tfrac{3}{2}m^2 + \tfrac{1}{2}\ell^2 - 50Nm - 26N\ell + 520N^2 + 42N^2\\ &\qquad = 2(m-11N)^2 + (20N-m-\ell)(6N+m) + \tfrac{1}{2}(20N-m-\ell)^2\end{aligned}$$

shows that this form is at least $-42N^2$ when $m, \ell \ge 0$ and $m+\ell \le 20N$. This gives the 2-adic contribution $\frac{21}{200}\log 2\cdot(20N)^2$.

Proposition 4.3. *There are positive integers D_N such that $D_N\Delta_N(i) \in \mathbb{Z}$ for every integer i, and, for every $\varepsilon > 0$ and all large N,*

$$\log D_N \le (\mathsf{A}+\varepsilon)(20N)^2, \qquad \mathsf{A} = \frac{8795}{4800} + \frac{21}{200}\log 2.$$

Proof. At an integer $z = i$ the local bounds above do not depend on i; at $p = 2$ the bound is weakest at $z = 0$. Let D_N be the product of the primes raised to these exponents. The estimate for $\log D_N$ is the sum of the contributions computed above. □

The polynomial Δ_N has degree at most $18N$ and takes values in $D_N^{-1}\mathbb{Z}$ at the integers. An integer-valued polynomial of degree d has coefficients in $\frac{1}{d!}\mathbb{Z}$, so

$$Q_N = (18N)!\, D_N\, \Delta_N(X) \in \mathbb{Z}[X], \qquad \deg Q_N \le 18N, \tag{4.3}$$

and $\log(18N)! = o(N^2)$.

4.5. **Nonvanishing.**

Proposition 4.4. *For all large primes p and every rational z such that pz has denominator prime to p, $\Delta_p(z) \neq 0$. In particular $Q_p(a/b) \neq 0$ whenever $a \in \mathbb{Z}$, $b \geq 1$ and $p^2 \nmid b$.*

Proof. For the present parameters OpenAI's reduction modulo p involves a rational base matrix B with 21 rows and 20 columns, and the three 20×20 matrices with rows $B_r + \sigma B_{r+1}$, for $\sigma \in \{0, 1, -1\}$, are invertible; this is checked by computing their inverses modulo the prime 1009, which shows that their determinants are nonzero rational numbers. OpenAI's argument then shows that the reduction of $p^2(\mathcal{A}_p^0 + z\mathcal{L}_p)$ modulo p is invertible for all large primes p when $p \nmid \operatorname{den} z$. If only pz has denominator prime to p, the difference $p^2 z\mathcal{L}_p$ from the matrix at $z = 0$ is still p-integral with zero reduction, because the numbers c_l have denominators that are powers of 2; so the reduction is again that of the matrix at $z = 0$. If $p^2 \nmid b$, then $p(a/b)$ has denominator prime to p. □

The computations modulo 1009 certify the base matrices; the conclusion for all large primes is a theorem and not an extrapolation from finitely many cases.

5. Divisibility of the coefficients

5.1. **The principle.** Proposition 4.3 uses, at each prime, only the poles of the individual entries. In the arithmetic method of Hata and Rhin–Viola one gains further by showing that all the coefficients of the linear forms share a large common factor. In our setting the analogue is the following. Let p be a prime and write $\mathcal{A}_{N,p}^0$ for the reduction modulo p of $p^2\mathcal{A}_N^0$, the matrix at the constant term $X = 0$. If its rank is at most $20N - e$, then by Lemma 4.2 the determinant $\Delta_N(z)$ gains a factor p^e at every rational z for which the reduction of $p^2(\mathcal{A}_N^0 + z\mathcal{L}_N)$ equals $\mathcal{A}_{N,p}^0$; in particular at every integer z.

A gain at the integers alone would only improve the denominator of $\Delta_N(a/b)$ for integer arguments. What we need is a gain in every coefficient of Q_N, because these coefficients are what the integer $b^{\deg Q_N} Q_N(a/b)$ is built from. For the primes considered below this follows from a second observation. Write $Q_N = \sum_j c_{N,j} X^j$, and let p^{40N-s} be the exact power of p in D_N. The matrix $p^2\mathcal{L}_N$ is p-integral, so

$$Q_N(X/p^2) = (18N)!\, \frac{D_N}{p^{40N}} \det(p^2\mathcal{A}_N^0 + X\,\mathcal{L}_N)$$

has coefficients divisible by p^{-s}, and hence $p^{2j-s} \mid c_{N,j}$. On the other hand, since $p^2\mathcal{L}_N \equiv 0 \pmod p$, the reduction of $p^2(\mathcal{A}_N^0 + z\mathcal{L}_N)$ at an integer z is $\mathcal{A}_{N,p}^0$, so a rank deficiency e gives $v_p(\Delta_N(z)) \geq e - 40N$ and $p^{e-s} \mid Q_N(z)$ for every integer z. In each of the intervals below $e \leq 2p$, so every coefficient with $j \geq p$ is divisible by p^{e-s}. The remaining part $\sum_{j<p} c_{N,j}X^j$ has degree less than p, and its values at $0, 1, \dots, p-1$ are divisible by p^{e-s}; since the Vandermonde matrix of these points is invertible modulo p, all its coefficients are divisible by p^{e-s}. Thus a rank deficiency e of the single reduced matrix $\mathcal{A}_{N,p}^0$ gives the common factor p^{e-s} of all the coefficients of Q_N. In the first and third intervals $s = 0$; in the second $s = (26N - 3p)_+$.

We prove three such rank deficiencies, for the primes in the intervals

$$\Big(\frac{52N}{9}, \frac{44N}{7}\Big), \qquad (8N, 9N), \qquad \Big(4N, \frac{9N}{2}\Big).$$

They are disjoint, and all their primes exceed $2N$. In each case the deficiency is proved by exhibiting explicit vectors in the left kernel of $\mathcal{A}_{N,p}^0$. The tools are the same throughout: the identities (4.1) for Chebyshev polynomials modulo p, the relation $(1-X)^p = 1 - X^p$, and a description of the residues of p^2 times the entries of the matrix in terms of the digits in base p of the coefficients of the row polynomials, which we now recall.

Fix p and write the raw columns (before the finite difference filter) as $j = Jp + \ell$ with $0 \leq \ell < p$. For a vector $c \in \mathbb{F}_p^{20N}$ put

$$P_c = \sum_r c_r P_r, \qquad D_c = \sum_r c_r D_r, \qquad W = (1 - X^2)^{(p-1)/2}.$$

Grouping OpenAI's formula for the entries by digits shows that the residue of p^2 times the raw entry in column j, applied to c, is a fixed linear combination of the coefficients $(XWP_c)[vp + \ell]$ and $D_c[vp + \ell]$, $v \geq 0$, with constants independent of c and ℓ. Here $F[k]$ denotes the coefficient of X^k in F. A vector c whose responses vanish on all raw columns in the range $[3N, 24N)$ lies in the left kernel of the filtered matrix.

5.2. **The primes in** $(52N/9, 44N/7)$**.** For all $N \geq 6310886697782126553600$ and all primes $52N/9 < p < 44N/7$,

(5.1)

$$\operatorname{rank} \mathcal{A}_{N,p}^0 \leq 20N - e_N(p), \qquad e_N(p) = \max\Big(0, \min\Big(\Big\lfloor \frac{9p - 52N}{2} \Big\rfloor, \Big\lfloor \frac{44N - 7p}{2} \Big\rfloor\Big)\Big),$$

The threshold on N guarantees that no prime of the interval divides the denominators of the finitely many rational constants used in the construction below.

Here the kernel vectors are built from rational differentials on the conic $x = (z + z^{-1})/2$, $s = (z - z^{-1})/2$. In this parametrization the source of each raw column is the coefficient of a Laurent expansion: with

$$\Phi(z) = \frac{z^{26N}(z-1)^{2N}}{(1+z^2)^{27N}},$$

the raw entry in column $j = Jp + \ell$ is, up to a fixed factor, the residue at $z = 0$ of $\Phi(z)\, x^{\ell} a_J(x)^p$, where a_J is a fixed polynomial expression in OpenAI's moment coefficients of the column digit J. A row combination is encoded by a differential $(U + sV)\, dz/z$, and its response to the digit J is the coefficient of x^5 in an explicit quadratic expression in U, V and the moment polynomials. We use a basis of six regular differentials and one differential with a double pole at $z = 0$, and five combinations of them whose responses to the digits $J = 0, \dots, 4$ are known exactly; the 5×7 response matrix has rational entries and a nonzero 4×4 minor $46085/1769472$.

For five polynomial inputs I_i of degree less than p we form, using the Frobenius map, the polynomial

$$K = (z-1)^p\, \mathcal{T} H_U + (z+1)^p\, \mathcal{T} H_V, \qquad H_U = \sum_i U_i^p I_i, \quad H_V = \sum_i V_i^p I_i,$$

where $\mathcal{T}$ is the substitution $x = (1+z^2)/(2z)$ cleared of denominators. Under a list of divisibility conditions on H_U and H_V at $x = 0$ and $x = \pm 1$, and one degree condition, K is divisible by a fixed polynomial T of large degree, and K/T has degree less than $20N$; its coefficients form a row combination c. The map from the inputs to c is injective: the involution $z \mapsto z^{-1}$ separates the two channels U and V, and the Frobenius map separates the coefficients of the inputs.

The responses of this row combination are computed directly from the identity

$$[x^{6p-1-\ell}](H(x)^p I(x)) = [x^5] H \cdot [x^{p-1-\ell}] I \qquad (\deg I < p),$$

which shows that the response at column $Jp + \ell$ is a fixed multiple of the coefficient of $x^{p-1-\ell}$ in the input. We take one input F with $(x-1)^{p-N} \mid F$, whose coefficients $[x^{p-1-\ell}]F$ are the values $\omega(\ell)$ of a polynomial ω of degree less than N; the finite difference of order N in the column filter annihilates them. The other inputs are chosen to satisfy the divisibility conditions at $x = 0$, and their responses fall on raw columns outside the range $[3N, 24N)$. Counting the dimension of the space of inputs against the number of linear conditions, separately for $p = 6N + \delta$ with $\delta > 0$ and for $p = 6N - d$, gives a kernel of dimension at least $N - (7\delta + 1)/2$ and $N - (9d + 1)/2$ respectively. These are the two expressions in (5.1).

5.3. **The primes in** $(8N, 9N)$**.**

Theorem 5.1. *Let $N \geq 1$ and let p be a prime with $8N < p < 9N$. Then the left kernel of $\mathcal{A}^0_{N,p}$ has dimension at least $18N - 2p - 1$.*

The row polynomials of our determinant are $P_r = (1-X)^N \operatorname{rev}(T_{|m|})$ and $D_r = \operatorname{sgn}(m)(1-X)^N \operatorname{rev}(U_{|m|-1})$, where $m = r - 2N$ and rev reverses the coefficients inside degree $26N - 1$. Write $\nu = (p+1)/2$. The first ingredient is a reflection formula for Chebyshev polynomials modulo p.

Lemma 5.2. *Let p be an odd prime. In $\mathbb{F}_p[X]$, for every integer m,*

$$T_{p-m} = X^p T_m - (X^2-1)^\nu U_{m-1}.$$

Proof. Both sides satisfy the three-term recurrence in m, so it suffices to check $m = 0$ and $m = 1$. This needs (4.1). The second identity there follows from the Pell equation $T_n^2 - (X^2-1)U_{n-1}^2 = 1$ at $n = p$, together with $(X^2-1)^p = X^{2p} - 1$; one cancels $X^2 - 1$ and compares leading coefficients. For $m = 1$ use $T_{p+1} = XT_p + (X^2-1)U_{p-1}$ and $T_{p-1} = 2XT_p - T_{p+1}$. □

Proof of Theorem 5.1. Reflecting the identity of Lemma 5.2 at degree $p + 26N$ gives, for each row,

$$(1-X^2)^\nu \operatorname{rev}(U_{m-1}) - X \operatorname{rev}(T_m) = -X^{p+1} \operatorname{rev}(T_{p-m}),$$

and the right side is divisible by X^{24N} because $|p - m| \le p + 2N$. Hence, for every c,

$$X^{24N} \mid (1-X^2)^\nu D_c - XP_c. \tag{5.2}$$

Let V be the space of c with $(1-X^2)^\nu \mid P_c$. Since $(1-X)^N$ divides every P_r, this is $\nu - N$ conditions at $X = 1$ and ν conditions at $X = -1$, that is $p+1-N$ linear conditions. For $c \in V$ write $P_c = (1-X^2)^\nu Y$ and $\widetilde{Q} = XY$. Since $(1-X^2)^p = 1 - X^{2p}$ in characteristic p,

$$XWP_c = (1 - X^{2p})\,\widetilde{Q}.$$

The polynomial P_c has no coefficients below X^{8N} and none from X^{27N} on, so $\widetilde{Q}$ is supported in degrees $[8N+1, 27N-p-1]$, and dividing (5.2) by the unit $(1-X^2)^\nu$ modulo X^{24N} gives $D_c[k] = \widetilde{Q}[k]$ for $k < 24N$.

Now let $27N - 3p \le \ell < 24N - 2p$. By the support of $\widetilde{Q}$, the base-p digits of XWP_c at the residue ℓ are $(0, \kappa, 0, -\kappa, 0, \ldots)$ with $\kappa = \widetilde{Q}[p+\ell]$, and those of D_c are $(0, \kappa, 0, 0, \ldots)$. So for $c \in V$ the response at every raw column $Jp + \ell$ in this range of residues is κ times a constant depending only on J: these columns have rank at most one on V.

For $p \in (8N, 9N)$ the raw columns $j \in [3N, 24N)$ with residue ℓ have $J \in \{1,2\}$ if $\ell < 27N - 3p$, $J \in \{0,1,2\}$ if $27N - 3p \le \ell < 24N - 2p$, and $J \in \{0,1\}$ if $24N - 2p \le \ell < p$. The number of linear conditions needed to make all responses vanish on V is therefore

$$2(27N - 3p) + (p - 3N) + 2(3p - 24N) = 3N + p.$$

Together with the $p + 1 - N$ conditions defining V this is $2p + 2N + 1$ conditions, and the space of solutions has dimension at least $20N - (2p + 2N + 1) = 18N - 2p - 1$. □

For these primes the normalizer has p-exponent $40N - (26N - 3p)_+$, so after the rescaling $X \mapsto X/p^2$ one obtains $p^e \mid c_{N,j}$ for all j with

$$e = e_N(p) = \min(p - 8N - 1,\ 18N - 2p - 1)_+.$$

5.4. **The primes in $(4N, 9N/2)$.**

Theorem 5.3. *Let $N \geq 1$ and let p be a prime with $4N < p$ and $2p < 9N$. Then the left kernel of $\mathcal{A}^0_{N,p}$ has dimension at least $(p - 4N - 1)/2$.*

Unlike Theorem 5.1, where the conditions are imposed on the row polynomials, here the argument concerns a block of columns. Put $d = p-4N$, which is odd, and $e = (d+1)/2$; the hypothesis $2p < 9N$ means $2d < N$. Write $S_c = \sum_r c_r \operatorname{rev}(T_{m_r})$ and $U_c = \sum_r c_r \operatorname{rev}(U_{m_r-1})$, so that $P_c = (1-X)^N S_c$, and put

$$\widehat{Q}_c = (X-1)^N XW\,(1-X)^N S_c, \qquad \widehat{D}_c = (X-1)^N (1-X)^N U_c.$$

Proof. An identity. Over any commutative ring, $U_{m-1}T_n - T_m U_{n-1} = U_{m-n-1}$, since both sides satisfy the recurrence in n and agree for $n = 0, 1$. In characteristic p take $m = 2p$ and use (4.1), $T_{2p} = 2X^{2p} - 1$ and $U_{2p-1} = 2T_p U_{p-1}$:

$$2X^p (X^2-1)^{(p-1)/2} T_n - (2X^{2p} - 1) U_{n-1} = U_{2p-n-1}.$$

Reflecting at degree $2p + 26N - 1$ and using $p \geq 4N$, the right side becomes divisible by X^{24N}. Summing over the rows with weights c_r and multiplying by $(X-1)^N(1-X)^N$ gives

$$2\,\widehat{Q}_c[i] = 2\,\widehat{D}_c[i] - \widehat{D}_c[i-2p] \qquad (i < 24N). \tag{5.3}$$

A block of columns. Let $M = [p - 7d, p - 6d)$ and consider the $5d$ filtered columns $k(J, \mu) = Jp + \mu - 4N$ for $1 \leq J \leq 5$ and $\mu \in M$. Each is the filter of order N applied to raw columns $Jp + (\mu - N + v)$, $0 \leq v \leq N$, which all have the same digit J because $N < \mu < p$ (this is where $2d < N$ is used). Since the alternating binomial sum $\sum_v (-1)^v \binom{N}{v} a_{m+v}$ is the coefficient of X^{m+N} in $(X-1)^N \sum_i a_i X^i$, the response of the column $k(J, \mu)$ is a fixed linear combination of the coefficients $\widehat{Q}_c[up + \mu]$ and $\widehat{D}_c[up + \mu]$.

Supports. The polynomials $\widehat{Q}_c$ and $\widehat{D}_c$ have no coefficients up to X^{8N}, and $\deg \widehat{Q}_c \leq 28N + p - 1$, $\deg \widehat{D}_c \leq 28N - 1$. For $\mu \in M$ only the digits $u = 2, \ldots, 6$ of $\widehat{Q}_c$ and $u = 2, \ldots, 5$ of $\widehat{D}_c$ can be nonzero, and for these $u \leq 5$ the relation (5.3) expresses $\widehat{D}_c[up + \mu]$ through $\widehat{Q}_c$. Hence the block columns vanish on c as soon as $\widehat{Q}_c[up + \mu] = 0$ for $2 \leq u \leq 6$ and $\mu \in M$.

A relation from the Frobenius map. Since $(1-X)^p = 1 - X^p$ and $2N + \frac{p-1}{2} + e = p$,

$$(1-X)^e\,\widehat{Q}_c = (1 - X^p)\,Y_c, \qquad Y_c = (-1)^N X (1+X)^{(p-1)/2} S_c.$$

Comparing coefficients at $X^{up+\mu}$ and summing over $u = 2, \ldots, 6$, the right side telescopes to $Y_c[6p + \mu] - Y_c[p + \mu]$, which vanishes for $\mu \geq p - 7d + e$. This gives $(d-1)/2$ linear relations among the numbers $\widehat{Q}_c[up + \mu]$, and they are triangular in μ.

Counting. Impose on c that the $20N - 5d$ filtered columns outside the block vanish, that $\widehat{Q}_c[up + \mu] = 0$ for $2 \leq u \leq 5$ and $\mu \in M$, and that $\widehat{Q}_c[6p + \mu] = 0$ for the first e values of μ. By the triangular relations the

remaining coefficients $\widehat{Q}_c[6p+\mu]$ vanish too, so every block column vanishes. This is $20N-5d+4d+e = 20N-(d-1)/2$ conditions, and the left kernel has dimension at least $(d-1)/2$. □

For these primes the normalizer has the full exponent $40N$, and the rank deficiency gives $p^{e_N(p)} \mid c_{N,j}$ for all j, with $e_N(p) = \lfloor (p-4N-1)/2 \rfloor$.

5.5. **The common factor.** Let Ω_N be the product of $p^{e_N(p)}$ over the primes of the three intervals. All coefficients of Q_N are divisible by Ω_N, and so is $Q_N(a/b)$ after multiplication by b^{18N}. The exponents are N times piecewise linear functions of p/N, up to $O(1)$: a function of area $16/63$ on $[52/9, 44/7]$, the function $\max(0, \min(x-8, 18-2x))$ of area $1/3$ on $[8,9]$, and $(x-4)/2$ of area $1/16$ on $[4, 9/2]$. By (4.2),

$$\log \Omega_N \geq (\chi - o(1))N^2, \qquad \chi = \frac{507}{2000} + \frac{3331}{10000} + \frac{624}{10000} = 0.649,$$

where each fraction is slightly less than the corresponding area. All the primes involved exceed $2N$, so the factor Ω_N does not interfere with the prime squares used in Section 7.

6. The real place

In this section we prove that Δ_N is small not only at G but on an exponentially small interval around it.

Theorem 6.1. *Put $\rho = 259299/2500$ and $\eta_\infty = 5993/3125$. For all large N,*

$$|Q_N(x)| \leq e^{-\eta_\infty N^2} \qquad \textit{whenever } |x-G| \leq e^{-\rho N}.$$

The proof has three parts. We first reduce the perturbation $\Delta_N(x) - \Delta_N(G)$ to determinants in which some columns of $\mathcal{A}_N$ are replaced by values of the row functions at fixed points (§6.1–6.2). We then bound such determinants by OpenAI's energy method, extended to configurations with fewer free points (§6.3). Finally we supply the trial sequences that the energy method needs and verify the resulting inequalities (§6.4–6.5).

6.1. **A quadrature formula for the coefficient of the constant.** Let $\delta = x - G$. Then $\Delta_N(x) = \det(\mathcal{A}_N + \delta \mathcal{L}_N)$, and the question is how large the columns of $\mathcal{L}_N$ are compared with those of $\mathcal{A}_N$. By (3.3) the entries of $\mathcal{L}_N$ are linear functionals of the coefficients of the row polynomials. Since these polynomials have bounded degree, the functionals can be written exactly as finite combinations of point evaluations: if $\tau_1, \dots, \tau_J$ are distinct points and J exceeds the degree of the polynomials involved, Lagrange interpolation gives

$$\mathcal{L}_N(r,k) = \sum_{a=1}^{J} A_r(\tau_a)\, \beta_{a,k}, \tag{6.1}$$

where the weights $\beta_{a,k}$ are the values of the same functionals on the Lagrange basis polynomials. This is a Gauss-type quadrature formula, exact

for polynomials of the relevant degree. The point of the formula is that the columns of $\mathcal{L}_N$ become combinations of point evaluations of the same row functions that define $\mathcal{A}_N$, so that the perturbed determinant can again be treated by the energy method. The cost is the size of the weights.

We take $J = 18N$ and the nodes

$$t_{j,\pm} = \pm\sqrt{a^2 + (1-a^2)(1+s_j)/2}, \qquad T_{9N}(s_j) = 0, \qquad a = \frac{2g_0}{1+g_0^2}, \quad g_0 = \frac{401}{2500},$$

that is, the roots of $P_N(t) = T_{9N}\big((2t^2 - (1+a^2))/(1-a^2)\big)$. They lie in $(-1,-a)\cup(a,1)$. In the coordinate x with $t = 2x/(1+x^2)$ every node has $|x| > g_0$; this matters below, because the energy fields are larger near $x = 0$.

Lemma 6.2. *After removal of the common factor of the rows, the weights in* (6.1) *satisfy*

$$|\beta_{a,k}| \le 108000N\, e^{17.913N}.$$

Proof. At a root, the derivative identity for Chebyshev polynomials gives $|P_N'(t_{j,\pm})| \ge 36Na/(1-a^2)$. The weight is a coefficient functional of the Lagrange polynomial, and every coefficient of a polynomial is bounded by its maximum modulus on a circle centered at the origin. On the circle $|z| = r$ with $r = 1001/1000$, write $s = |\operatorname{Re} z|$. For $w = (2z^2 - (1+a^2))/(1-a^2)$ one has $|T_{9N}(w)| \le (\Theta + \sqrt{\Theta^2-1})^{9N}$, where

$$\Theta(s) = \frac{\sqrt{(r^2+1)^2 - 4s^2} + \sqrt{(r^2+a^2)^2 - 4a^2s^2}}{1-a^2}$$

is half the sum of the distances from w to ±1. The function Θ decreases in s, while the factor $|1-z|^2 \le r^2 + 1 + 2s$ coming from the column filter increases. A cover of $[0,r]$ by 55 intervals, with rational upper bounds for the square roots and logarithms at their endpoints, proves

$$9\log\big(\Theta(s) + \sqrt{\Theta(s)^2-1}\big) + \log(r^2+1+2s) < 17.913 \qquad (0 \le s \le r).$$

Since $|z - t_{j,\pm}| \ge 1/1000$ on the circle and $(1-a^2)/(36a) \le 1$, the bound follows. □

The circle of radius 1.001 instead of 1 separates the circle from the nodes; on the unit circle itself the cancellation in the factor $(1-t)^N$ near $t = 1$ is what keeps the bound small.

6.2. **Column expansion and an identity of Andréief type.** Expanding $\det(\mathcal{A}_N + \delta\Phi\beta)$, where $\Phi(r,a) = A_r(\tau_a)$, multilinearly in the columns gives a sum over maps f from the columns to $\{\varnothing, 1, \dots, J\}$ of $\prod_{f(k)\neq\varnothing} \delta\beta_{f(k),k} \det M_f$. Here M_f is the matrix whose k-th column is the k-th column of $\mathcal{A}_N$ if $f(k) = \varnothing$ and the vector $(A_r(\tau_{f(k)}))_r$ otherwise.

Lemma 6.3. *If $|\beta_{a,k}| \le B$ and $|\det M_f| \le Mc^{m(f)}$ for every f, where $m(f)$ is the number of columns replaced, then*

$$\big|\det(\mathcal{A}_N + \delta\Phi\beta) - \det\mathcal{A}_N\big| \le M\big((1 + J|\delta|Bc)^n - 1\big).$$

Each M_f has $p = n - m$ columns that are integrals and m columns that are point values. Andréief's identity extends to this situation.

Lemma 6.4. *Let μ, ν be finite measures, F_r $(r < p + m)$, G_i $(i < p)$ and K integrable functions, and let B be the matrix with $B(r,i) = \iint F_r(t)K(t,s)G_i(s)\,d\mu(t)\,d\nu(s)$ for $i < p$ and $B(r, p+j) = F_r(\tau'_j)$. Then*

$$(p!)^2 \det B = \iint \det\big[F_r((t\|\tau')_c)\big]_{r,c} \det[K(t_i,s_j)]_{i,j<p} \det[G_i(s_j)]_{j,i<p}\, d\mu^p(t)\, d\nu^p(s),$$

where $t\|\tau'$ is the list of the p variables followed by the m fixed points.

Proof. For $m = 0$ this is Andréief's identity applied twice [11]. In general expand $\det B$ multilinearly in its first p columns, which introduces one pair of variables (t_i, s_i) for each of them, and antisymmetrize first in the t and then in the s variables; the last m columns are constant throughout. □

With $K(t,s) = 1/(1-ts)$ the middle determinant is the Cauchy determinant, and the third is a generalized Vandermonde determinant $\prod_j s_j^b (1-s_j)^q \det[s_j^{E_i}]$. For the latter we use the bound

$$|\det[s_j^{E_i}]| \le \Big(\prod_j (1+s_j)\Big)^{n-p} \prod_{i<j} |s_i - s_j|, \tag{6.2}$$

valid for distinct exponents $E_i < n$ and $s_j \in [0,1]$. It follows by multiplying the matrix by a unitriangular matrix built from the polynomial $\prod_j (X - s_j)$ and bounding the remaining $(n-p)\times(n-p)$ determinant by the product of the ℓ^1 norms of its columns. The quotient $\det[s_j^{E_i}]/\prod_{i<j}(s_j - s_i)$ is a Schur polynomial, so the bound is close to optimal.

6.3. **The energy estimate with fixed points.** OpenAI bounds the integrand in Andréief's identity by the exponential of $n^2 E$, where E is a discrete logarithmic energy of the points x_i and s_j in the coordinate $t = 2x/(1+x^2)$. The energy consists of an external field for each point, logarithmic interactions $\log|x_i - x_j|$, $\log(1 - x_i x_j)$ and $\log|s_i - s_j|$ among points of the same kind, and an interaction $-\log(1 - 2x_i s_j + x_i^2)$ between the two kinds. There are two cases, according to which of OpenAI's two bounds for the determinant of the row functions is used [25, Props. 5.2, 5.3]; in the second case the interaction among the x_i has weight 2 instead of 1.

The energy is bounded from above by duality. Given trial sequences $u = (u_k)$ and $v = (v_k)$, one writes the interaction terms as quadratic forms in the moments $\frac{1}{n}\sum_i T_k(x_i)$ and $\frac{1}{n}\sum_j s_j^k$ and completes the squares against u and v. What remains is the sum of the norms $\|u\|^2$, $\|v\|^2$ and, for each point, a function of one variable,

$$X(x) = W_x(x) - 2\kappa\, T(u,x) - S(v,x), \qquad Y(s) = W_s(s) + 2\, T(v,s),$$

where W_x, W_s are the external fields, $\kappa \in \{1,2\}$ is the case, and $T(u,x) = \sum_k u_k T_k(x)/k$, $S(v,x) = \sum_k v_k x^k/k$ are the potentials of the trial sequences.

If $X \le M_X$ and $Y \le M_Y$ on the whole domains, the energy of any configuration is at most a constant determined by M_X, M_Y and the norms. In potential-theoretic terms the extremal configuration is described by an equilibrium measure for two interacting measures in an external field [31], and the best trial sequences are its moments, as Carreira observed [6]; any trial sequences give a valid upper bound.

When m of the n columns are point values, the configuration has only $p = n-m$ free points of each kind, while the normalization is still by n. The fixed points contribute through the determinant of the row functions evaluated at the concatenated list $t\|\tau'$. One checks that every step of OpenAI's argument remains valid uniformly in $p \le n$: the regularization of the logarithmic interactions, the diagonal corrections, and the completion of squares only involve sums over the points present. The missing points change the energy bound by an amount linear in m, which is the cost of a fixed column.

For the parameters of Section 4 this cost is controlled as follows. We use two pairs of trial sequences, one for each case. For each case the cost of a fixed column is at most $20c_\kappa N$ in the exponent, where c_κ is obtained by completing a square in the proportions $\alpha = m/n$ and $\tau \le \alpha$ of fixed points of the two kinds. For the second case the quantity to be bounded is

$$h_0 + (2.6\log 2 - F + R - c_2)\alpha - 2\log 2\,\alpha^2 + (\log 2 - R - 0.75)\tau - \log 2\,\tau^2, \tag{6.3}$$

with $0 \le \tau \le \alpha \le 1$, where F is the bound for the sum of the two fields, R is the bound for the field at the fixed points, and h_0 collects the constants. Here the fixed points are the quadrature nodes, at which $|x| > g_0$; this is why the bound R for the field at the fixed points is needed only on $g_0 \le |x| \le 1$, where it is much better than on the whole interval. The expression (6.3) is negative on its domain when $c_2 = 4.29033$, and the analogous expression for the first case is negative when $c_1 = 4.21$.

Combining this with Lemma 6.2, a determinant M_f with m fixed columns contributes at most $(540000N)^m e^{(20c_2+17.913)Nm}$ times the unperturbed bound; the factor 5^m comes from expanding the row functions at the fixed points in the second case. Lemma 6.3 then gives, for $|\delta| \le e^{-\rho N}$ with $\rho = 20c_2 + 17.913 = 259299/2500$, a factor $(1 + O(N^2))^{20N} = e^{o(N^2)}$. This proves Theorem 6.1, given the numerical bounds of the next two subsections; the constant η_∞ comes from

$$400\big(U_* - \tfrac{1}{2}\log 2 - \mathsf{A}\big) = 400U_* - \frac{8795}{12} - 242\log 2 > \frac{5993}{3125},$$

where U_* is the energy bound at the real place given below, $\frac{1}{2}\log 2$ is the normalization of the determinant, and A is the constant of Proposition 4.3.

6.4. **The trial sequences.** For the first case we use a trial sequence with 40 Chebyshev coefficients and a finite number of geometric terms, with an additional multiplier $\lambda = 1.68493357$ for the constraint that defines the case.

For the second case we use trial sequences of the form

$$u_k = c_k + \operatorname{Re}\sum_j r_j z_j^k \qquad (k \ge 1;\ c_k = 0 \text{ for } k > K), \tag{6.4}$$

with $K = 640$ Chebyshev coefficients in $2^{-40}\mathbb{Z}$ and 56 geometric terms, 22 of them with complex z_j and complex r_j.

The geometric terms have a simple meaning. A term (z, r) contributes $-\frac{1}{2}\operatorname{Re} r\log(1 - 2xz + z^2)$ to the potential $T(u, x)$. If $x_0 + i\epsilon = \frac{1}{2}(z + z^{-1})$ then $1 - 2xz + z^2 = 2z(x_0 + i\epsilon - x)$, so the term is, up to a constant, the logarithmic potential of a point charge of complex strength r at $x_0 + i\epsilon$ and its conjugate. The extremal measure of the energy problem has a gap in its support around $x = 0$, of radius about 0.1, and its potential has a logarithmic singularity at $x = 0$ and square-root behaviour at the edges of the gap. A Chebyshev expansion converges slowly near such points. A charge at a small distance ϵ from them reproduces the local behaviour with one term. We place the complex charges at $x_0 \in \{0, \pm 0.03, \pm 0.06, \pm 0.09, \pm 0.12, \pm 0.16\}$ with $\epsilon \in \{10^{-2}, 3 \cdot 10^{-3}\}$; the numbers z_j are rationals with twelve decimals. The norms of the geometric terms have closed forms: two terms (z_j, r_j), (z_l, r_l) contribute $-\frac{1}{2}\operatorname{Re}\bigl(r_j r_l \log(1 - z_j z_l) + r_j \bar{r}_l \log(1 - z_j \bar{z}_l)\bigr)$.

Without the complex terms, the same 640 coefficients and the 34 real geometric terms give, in a numerical optimization, no exponent below 728.5.

6.5. **Verification of the bounds.** We must bound three functions of one variable from above on their whole domains: the field X on $[-1, 1]$, the field Y on $[0, 1]$, and the field at the fixed points on $g_0 \le |x| \le 1$. With $x = \cos\theta$, each is a constant plus a cosine series with 640 terms plus finitely many terms $\operatorname{Re} w\log(1 - ae^{im\theta})$.

We cover each domain by intervals $[x_c - h, x_c + h]$ whose endpoints are multiples of 2^{-40}, and on each interval bound the function by its Taylor expansion at x_c. The polynomial part has degree 639, and a global bound for its fifth derivative would force very short intervals. Instead we compute the derivatives of the polynomial part exactly at x_c up to order 70, and bound only the remainder beyond order 70 by a single global constant, which enters multiplied by $h^{70}/70!$. The logarithmic terms are expanded to order 5, with remainders bounded from the exact minimum of $|1 - ae^{im\theta}|$ on the interval. A few intervals at $x = 0$ and $x = \pm 1$ treat the singular terms separately. The covers have 598, 435 and 366 intervals. Each interval is checked by exact rational and integer arithmetic.

The resulting bounds are

$$\sup X \le -\frac{97745957}{10^8}, \qquad \sup Y \le -\frac{156655368}{10^8}, \qquad \sup_{g_0 \le |x| \le 1} R \le -\frac{5247033}{10^8},$$

the norms satisfy $2\|u\|^2 + \frac{1}{2}\|v\|^2 \le \frac{24149189}{31250000}$, and the first coefficient satisfies $u_1 \ge 9/50$, which is used near $x = 0$. These give the energy bound $U_* = 141027507/62500000$ at the real place.

The trial data are fixed once and for all; the file containing them has SHA-256 digest

`6a055ec808510b5ff836a819aebc837ea1593f3c96263318ac2688a59659af02`.

7. Proof of Theorem 1.1

7.1. **The usual criterion.** We first recall the standard way of passing from the three properties of the polynomials Q_N to a measure, to explain where the remaining ideas enter.

Proposition 7.1. *Let $\xi \in \mathbb{R}$, $d \in \mathbb{N}$ and $\eta, \rho > 0$, and let $Q_p \in \mathbb{Z}[X]$ be polynomials, indexed by primes p, such that for all large p: $\deg Q_p \le dp$; $Q_p(a/b) \neq 0$ whenever $p \nmid b$; and $|Q_p(x)| \le e^{-\eta p^2}$ whenever $|x - \xi| \le e^{-\rho p}$. Then for every $\varepsilon > 0$, $\mu(\xi) \le (1+\varepsilon)\rho \max(d/\eta, c/\varepsilon)$ for every $c > 1$.*

Proof. Let $|\xi - a/b|$ be small and $X = \max(d/\eta, c/\varepsilon)\log b$. By the prime number theorem $\theta((1+\varepsilon)X) - \theta(X) = \varepsilon X + o(X) > \log b$, so some prime $p \in (X, (1+\varepsilon)X]$ does not divide b. If $|\xi - a/b| \le b^{-(1+\varepsilon)\rho\max(d/\eta, c/\varepsilon)}$, then $|\xi - a/b| \le e^{-\rho p}$ and the nonzero integer $b^{dp}Q_p(a/b)$ has absolute value at most $b^{dp}e^{-\eta p^2} < 1$, since $p > X \ge (d/\eta)\log b$. □

The loss $\max(d/\eta, c/\varepsilon)$ instead of d/η is the cost of finding a prime that does not divide b in a short interval. We now remove it, using the two remaining properties of Q_N: the nonvanishing under $p^2 \nmid b$ (Proposition 4.4) and the divisibility of the coefficients by prime squares.

7.2. **Prime squares in the coefficients.**

Lemma 7.2. *For all large N and every prime $N < q \le 2N$, the j-th coefficient of Q_N is divisible by q^{2j}.*

Proof. For these primes the matrices $q^2\mathcal{A}_N^0$ and $\mathcal{L}_N$ are q-integral, and q^{40N} divides D_N. The identity

$$Q_N(X/q^2) = (18N)!\,\frac{D_N}{q^{40N}}\det(q^2\mathcal{A}_N^0 + X\mathcal{L}_N)$$

shows that $Q_N(X/q^2)$ has q-integral coefficients; its j-th coefficient is $c_{N,j}/q^{2j}$. □

Consequently, if B is a product of squares q^2 of distinct primes $N < q \le 2N$ with $B \mid b$, then every term $c_{N,j}a^j b^{18N-j}$ of $b^{18N}Q_N(a/b)$ is divisible by B^{18N}, and

$$1 \le \Big(\frac{b}{B}\Big)^{18N}\Big|\frac{Q_N(a/b)}{\Omega_N}\Big| \qquad \text{whenever } Q_N(a/b) \neq 0. \tag{7.1}$$

Here we have also divided by the common factor Ω_N of Section 5, whose primes exceed $2N$ and therefore do not meet B.

7.3. **The choice of the prime.** Put

$$\rho = \frac{259299}{2500}, \qquad \eta = \eta_\infty + \chi = \frac{5993}{3125} + \frac{649}{1000} = \frac{64169}{25000}, \qquad k = \frac{728}{\rho} = \frac{1820000}{259299}, \qquad \delta_0 = 10^{-7}.$$

By Theorem 6.1 and Section 5, the integer polynomials $\widetilde{Q}_N = Q_N/\Omega_N$ satisfy $|\widetilde{Q}_N(x)| \le e^{-(\eta-o(1))N^2}$ for $|x-G| \le e^{-\rho N}$; their degree is at most $18N$, and they do not vanish at a/b for $N = p$ prime with $p^2 \nmid b$.

Let a/b be a rational number with b large, put $L = \log b$, $U = kL$ and $V = (k-1)L$, and suppose $|G - a/b| \le b^{-728}$. By the prime number theorem, $(1-\delta_0)t \le \theta(t) \le (1+\delta_0)t$ for all large t. If every prime $p \in (V, U]$ had $p^2 \mid b$, then $2(\theta(U) - \theta(V)) \le L$; but

$$2\big(\theta(U) - \theta(V)\big) \ge 2L\big(1 - \delta_0(2k-1)\big) > L.$$

Let p be the *largest* prime in $(V, U]$ with $p^2 \nmid b$. Every prime $q \in (p, U]$ has $q^2 \mid b$, and $q \le U < 2V < 2p$ because $k > 2$. Let B be the product of these q^2, so that $\log B = 2T$ with $T = \theta(U) - \theta(p) \ge U - p - 2\delta_0 U$.

Since $\rho p \le \rho U = 728L$, we have $|a/b - G| \le e^{-\rho p}$, and (7.1) with $N = p$ gives $\eta p^2 \le 18p(L - 2T) + o(p^2)$. Hence

$$\eta p \le 18\big(L - 2U + 2p + 4\delta_0 U\big) + o(p).$$

As $\eta < 36$, the left side minus $36p$ decreases in p, and $p \le U = kL$ gives

$$\eta k \le 18 + 72\delta_0 k + o(1).$$

But

$$\eta k - 18 - 72\delta_0 k = \frac{513512}{32412375} > 0,$$

a contradiction. This proves Theorem 1.1.

The choice of the largest admissible prime is what makes the selection free: the primes of the interval that are not admissible have their squares in b, and these squares are recovered from the coefficients through Lemma 7.2. In the limit $\delta_0 \to 0$ the argument gives the exponent

$$\frac{18\rho}{\eta} = \frac{46673820}{64169} = 727.357\ldots.$$

8. Limits of the method

The exponent is determined by three numbers: the degree 18, the margin η and the perturbation exponent ρ. The margin is the sum of the difference between the real and the finite places and the contribution χ of the common factor. The perturbation exponent is the cost $20c_2$ of a fixed column plus the logarithmic size 17.913 of the quadrature weights.

Within the framework of Section 6, with the gap radius $g_0 = 401/2500$, the weight 17.913 and the exponent $\chi = 0.649$ of the common factor fixed, the integer 728 is close to optimal. A weak dual certificate on 934 rational intervals, verified in interval arithmetic at 60 and 80 digits, shows that no pair of trial sequences gives an integer exponent $B \le 726$ in this framework; its margin is $3.5 \cdot 10^{-6}$ at $B = 726$ and $-5.3 \cdot 10^{-6}$ at $B = 727$, and finer

subdivisions approach 726.5. This computation is not part of the formal proof, and it says nothing about $\mu(G)$ itself. Further progress requires new arithmetic, which enlarges χ or reduces the degree, or a better treatment of the fixed columns, which reduces ρ.

9. Formalization

All results of this paper are formalized in Lean 4 [7] with Mathlib [34], on top of OpenAI's formalization of the irrationality of G [25]. The development is available at

`https://github.com/slitsey/catalan-irrationality-measure`.

The statement of Theorem 1.1 in `Challenge.lean` uses only Mathlib, with Catalan's constant written as the series $\sum_j (-1)^j/(2j+1)^2$; its proof is `Solution.lean`. The solution has no `sorry` and no `native_decide`, and the command `#print axioms` reports only the three standard axioms `propext`, `Classical.choice` and `Quot.sound`. The file `comparator.json` is a configuration for `lake comparator` [20] stating that `Solution` proves the statement of `Challenge` with these axioms only. We have not run Comparator, which requires Linux; the axiom check above was done with `#print axioms`. The prime number theorem is taken from the PNT+ project [35].

The proof combines conventional formal proofs with numerical certificates checked by the Lean kernel: the inverses of the base matrices modulo 1009, the response matrices of Section 5.2, the cover of Lemma 6.2, the trial sequences of Section 6.4 and the covers of Section 6.5. All are exact rational or integer computations, decided by the kernel without compiled code. Large certificates are split into many small declarations with explicit intermediate values. Where a recurrence produces integers divisible by large powers of 2, as the Chebyshev and Clenshaw recurrences do, the certificates run on the quotients by these powers, with one lemma relating the two. The interval arithmetic used to generate the trial data and the covers [16] is not trusted: Lean rechecks every inequality.

The parts of OpenAI's formalization that are used were converted to Lean's module system and ported to Lean `v4.35.0-rc2`; the repository's `README` and `PORTING.md` describe the changes.

Disclosure of AI use

The mathematics, the Lean formalization, the numerical computations and certificate generators, the literature search and the text of this paper were produced by AI systems at the request of the author, who posed the problem, directed the work, and is responsible for this paper. Claude Opus 5.5 and Claude Sonnet 5.5 (Anthropic) produced the measure framework and the perturbation estimate by quadrature (Section 6), the rank deficiencies of Theorems 5.1 and 5.3 with their consequences for the coefficients, the trial sequences with complex terms and the verification of

Section 6.5, the final assembly, and the corresponding Lean proofs. OpenAI Codex produced the determinant of size $20N$ with its local arithmetic and nonvanishing, the energy estimates with fixed columns and the fields for the first case, the quadrature nodes avoiding the gap and the bound of Lemma 6.2, the divisibility of the coefficients by prime squares and the choice of the largest admissible prime, the rank deficiency of Section 5.2, and the corresponding Lean proofs. The underlying irrationality proof and its formalization are OpenAI's [25].

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